

N. Shree Deepthi Batch – 37

AI-ASSISTED CODING ASSIGNMENT 14

Task 1: AI-

Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:

- o Home
- o About
- o Projects
- o Contact

2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.

3. Customize AI-generated CSS to add:

- o Hover effects on project cards
- o Smooth scrolling navigation

Prompt

Generate a responsive personal portfolio website using HTML5, CSS3, and JavaScript.

- Use semantic HTML structure with sections: Home, About, Projects, and Contact.
- Create a navigation bar that scrolls smoothly to each section.
- Design the Projects section using CSS Flexbox or Grid with project cards.
- Add hover effects to the project cards (scale effect or shadow).
- Ensure the layout is fully responsive for desktop, tablet, and mobile screens.
- Implement smooth scrolling navigation using JavaScript.
- Use modern styling with clean typography and spacing.

Code:

Index.html

```
index.html X # styles.css
index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Personal Portfolio</title>
7      <link rel="stylesheet" href="styles.css">
8  </head>
9  <body>
10     <nav id="navbar">
11         <div class="nav-container">
12             <h1 class="logo">My Portfolio</h1>
13             <ul class="nav-links">
14                 <li><a href="#home">Home</a></li>
15                 <li><a href="#about">About</a></li>
16                 <li><a href="#projects">Projects</a></li>
17                 <li><a href="#contact">Contact</a></li>
18             </ul>
19         </div>
20     </nav>
21
22     <section id="home">
23         <div class="hero">
24             <h1>Welcome to My Portfolio</h1>
25             <p>I'm a passionate developer creating amazing web experiences.</p>
26             <a href="#about" class="btn">Learn More</a>
27         </div>
28     </section>
29
```

```
index.html X # styles.css
> index.html > ...
 2  <html lang="en">
 9  <body>

30  <section id="about">
31    <div class="container">
32      <h2>About Me</h2>
33      <div class="about-content">
34        <div class="about-text">
35          <p>Hello! I'm a web developer with a passion for creating beautiful
36          <p>With experience in HTML, CSS, and JavaScript, I love bringing i
37        </div>
38        <div class="about-image">
39          
41      </div>
42    </div>
43  </section>
44
45  <section id="projects">
46    <div class="container">
47      <h2>My Projects</h2>
48      <div class="projects-grid">
49        <div class="project-card">
50          <h3>Project 1</h3>
51          <p>A responsive web application built with React and Node.js.</p>
52          <a href="#" class="btn">View Project</a>
53        </div>
54        <div class="project-card">
55          <h3>Project 2</h3>
56          <p>An e-commerce platform with modern UI/UX design.</p>
```

```

57        <div class="project-card">
58          <h3>Project 3</h3>
59          <p>A portfolio website showcasing creative designs.</p>
60          <a href="#" class="btn">View Project</a>
61        </div>
62        <div class="project-card">
63          <h3>Project 4</h3>
64          <p>A mobile-first blog application with CMS integration.</p>
65          <a href="#" class="btn">View Project</a>
66        </div>
67      </div>
68    </div>
69  </section>
```

```

<section id="contact">
  <div class="container">
    <h2>Contact Me</h2>
    <div class="contact-content">
      <div class="contact-info">
        <p>Let's work together! Feel free to reach out.</p>
        <p>Email: your.email@example.com</p>
        <p>Phone: (123) 456-7890</p>
      </div>
      <form class="contact-form">
        <input type="text" placeholder="Your Name" required>
        <input type="email" placeholder="Your Email" required>
        <textarea placeholder="Your Message" required></textarea>
        <button type="submit" class="btn">Send Message</button>
      </form>
    </div>
  </div>
</section>

<script src="script.js"></script>
</body>
</html>

```

```

index.html  # styles.css  X
styles.css > ...
1  /* Reset and Base Styles */
2  * {
3    margin: 0;
4    padding: 0;
5    box-sizing: border-box;
6  }
7
8  body {
9    font-family: 'Arial', sans-serif;
10   line-height: 1.6;
11   color: #333;
12   scroll-behavior: smooth;
13 }
14
15 .container {
16   max-width: 1200px;
17   margin: 0 auto;
18   padding: 0 20px;
19 }
20
21 /* Navigation */
22 #nav {
23   position: fixed;
24   top: 0;
25   width: 100%;
26   background-color: #fff;
27   box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
28   z-index: 1000;
29 }

```

Style.css

```

31 .nav-container {
32     display: flex;
33     justify-content: space-between;
34     align-items: center;
35     padding: 1rem 2rem;
36 }
37
38 .logo {
39     font-size: 1.5rem;
40     font-weight: bold;
41 }
42
43 .nav-links {
44     display: flex;
45     list-style: none;
46 }
47
48 .nav-links li {
49     margin-left: 2rem;
50 }
51
52 .nav-links a {
53     text-decoration: none;
54     color: #333;
55     font-weight: 500;
56     transition: color 0.3s;
57 }
58
59 .nav-links a:hover {

```

```

3  /* Sections */
4  section {
5      padding: 5rem 0;
6      min-height: 100vh;
7      display: flex;
8      align-items: center;
9  }
10
11 #home {
12     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
13     color: white;
14 }
15
16 .hero {
17     text-align: center;
18     max-width: 800px;
19     margin: 0 auto;
20 }
21
22 .hero h1 {
23     font-size: 3rem;
24     margin-bottom: 1rem;
25 }
26
27 .hero p {
28     font-size: 1.2rem;
29     margin-bottom: 2rem;
30 }

```

```

92  .btn {
93      display: inline-block;
94      padding: 0.8rem 2rem;
95      background-color: #007bff;
96      color: white;
97      text-decoration: none;
98      border-radius: 5px;
99      transition: background-color 0.3s, transform 0.3s;
100 }
101
102 .btn:hover {
103     background-color: #0056b3;
104     transform: translateY(-2px);
105 }
106
107 /* About Section */
108 .about-content {
109     display: flex;
110     align-items: center;
111     gap: 2rem;
112 }
113
114 .about-text {
115     flex: 1;
116 }
117
118 .about-image {
119     flex: 1;
120     text-align: center;

```

```

123 .about-image img {
124     width: 300px;
125     height: 300px;
126     border-radius: 50%;
127     object-fit: cover;
128 }
129
130 /* Projects Section */
131 .projects-grid {
132     display: grid;
133     grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
134     gap: 2rem;
135 }
136
137 .project-card {
138     background-color: #f8f9fa;
139     padding: 2rem;
140     border-radius: 10px;
141     box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
142     transition: transform 0.3s, box-shadow 0.3s;
143 }
144
145 .project-card:hover {
146     transform: translateY(-5px);
147     box-shadow: 0 8px 15px rgba(0, 0, 0, 0.2);
148 }
149
150 .project-card h3 {
151     margin-bottom: 1rem;

```

```

183 .contact-form textarea {
184     resize: vertical;
185     min-height: 150px;
186 }
187
188 /* Responsive Design */
189 @media (max-width: 768px) {
190     .nav-container {
191         flex-direction: column;
192         padding: 1rem;
193     }
194
195     .nav-links {
196         margin-top: 1rem;
197     }
198
199     .nav-links li {
200         margin-left: 0;
201         margin-right: 1rem;
202     }
203
204     .hero h1 {
205         font-size: 2rem;
206     }
207
208     .about-content {
209         flex-direction: column;
210         text-align: center;
211     }

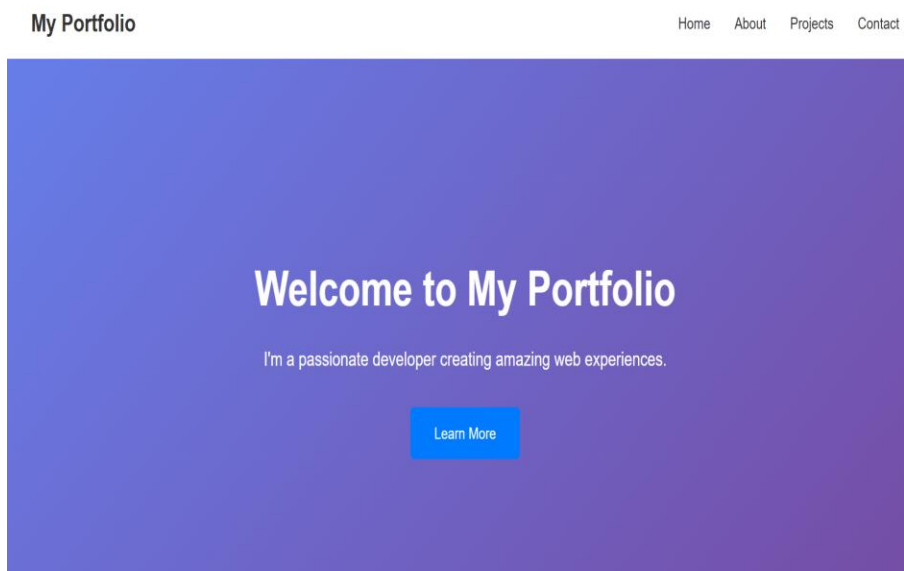
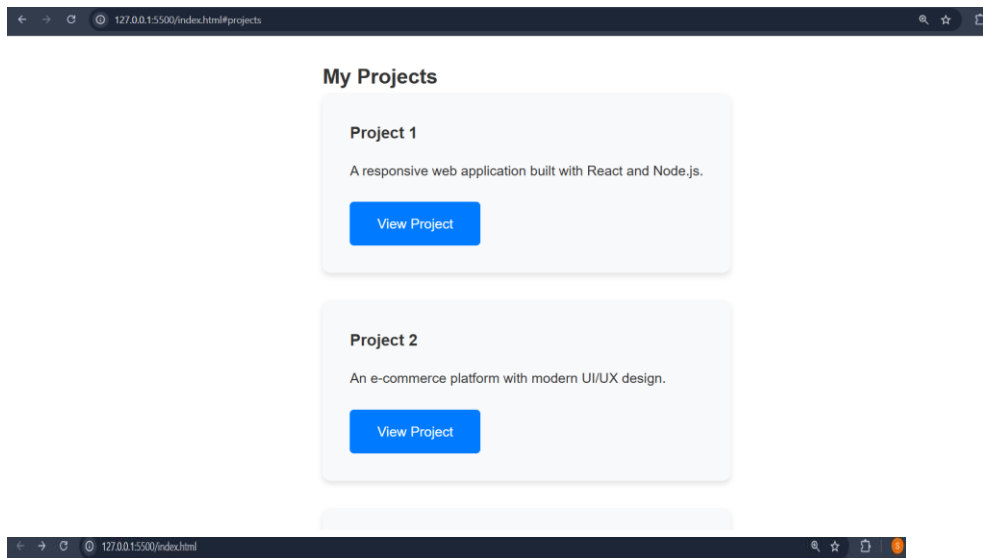
```

```

# styles.css > ...
154 .project-card p {
155     margin-bottom: 1.5rem;
156 }
157
158 /* Contact Section */
159 .contact-content {
160     display: flex;
161     gap: 2rem;
162 }
163
164 .contact-info {
165     flex: 1;
166 }
167
168 .contact-form {
169     flex: 1;
170     display: flex;
171     flex-direction: column;
172 }
173
174 .contact-form input,
175 .contact-form textarea {
176     margin-bottom: 1rem;
177     padding: 0.8rem;
178     border: 1px solid #ddd;
179     border-radius: 5px;
180     font-size: 1rem;
181 }
182

```

Output:



Contact Me

Let's work together! Feel free to reach out.

Email: your.email@example.com

Phone: (123) 456-7890

Code Explanation

The personal portfolio website is built using **HTML, CSS, and JavaScript** to showcase a student's profile, skills, projects, and contact details.

HTML Structure

The HTML file creates the structure of the website using **semantic tags**.

- The <header> section contains the **navigation bar** with links to different sections such as **Home, About, Projects, and Contact**.
- Each section is created using <section> tags with unique IDs so that navigation links can scroll to the correct part of the page.
- The **Projects section** displays project cards that contain project title and description.
- The **Contact section** includes contact information or a form.

CSS Styling

CSS is used to improve the visual appearance of the portfolio.

- **Flexbox or Grid** is used to arrange project cards in a structured layout.
- Hover effects are added to project cards using the :hover selector to create animations such as **scaling or shadow effects**.
- Media queries are used to make the website **responsive**, ensuring it adapts properly to **desktop, tablet, and mobile screens**.

JavaScript Functionality

JavaScript is used to enhance user interaction.

- **Smooth scrolling** is implemented so that when users click navigation links, the page smoothly scrolls to the corresponding section.
- Event listeners detect navigation clicks and trigger scrolling behavior.

This results in a **responsive and interactive personal portfolio website**.

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its

offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

- o Home

- o Menu

- o Gallery

- o Contact

2. Generate a menu section styled with CSS cards.

3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.

4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

Prompt used:

Create a responsive restaurant landing page using HTML, CSS, and JavaScript.

- Add a navigation bar with links: Home, Menu, Gallery, Contact.
- Create a hero section with restaurant name and tagline.
- Design a Menu section using CSS cards to display food items with name, image, and price.
- Create a Gallery section with an image slider using JavaScript.
- The image slider should have automatic sliding and next/previous controls.
- Add smooth hover animations on menu cards.
- Ensure the page is responsive using Flexbox or CSS Grid.

AI Generated Code:

Index.html

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Delicious Bites Restaurant</title>
7   <link rel="stylesheet" href="restaurant.css">
8 </head>
9 <body>
10   <nav id="navbar">
11     <div class="nav-container">
12       <h1 class="logo">Delicious Bites</h1>
13       <ul class="nav-links">
14         <li><a href="#home">Home</a></li>
15         <li><a href="#menu">Menu</a></li>
16         <li><a href="#gallery">Gallery</a></li>
17         <li><a href="#contact">Contact</a></li>
18       </ul>
19     </div>
20   </nav>
21
22   <section id="home" class="hero">
23     <div class="hero-content">
24       <h1>Welcome to Delicious Bites</h1>
25       <p>Experience the finest cuisine in a cozy atmosphere</p>
26       <a href="#menu" class="btn">View Our Menu</a>
27     </div>
28   </section>

```

```

<section id="menu">
  <div class="container">
    <h2>Our Menu</h2>
    <div class="menu-grid">
      <div class="menu-card">
        
        <h3>Grilled Salmon</h3>
        <p>Fresh Atlantic salmon grilled to perfection</p>
        <span class="price">$24.99</span>
      </div>
      <div class="menu-card">
        
        <h3>Beef Steak</h3>
        <p>Juicy ribeye steak with garlic butter</p>
        <span class="price">$32.99</span>
      </div>
      <div class="menu-card">
        
        <h3>Pasta Carbonara</h3>
        <p>Classic Italian pasta with pancetta and eggs</p>
        <span class="price">$18.99</span>
      </div>
      <div class="menu-card">
        
        <h3>Caesar Salad</h3>
        <p>Crisp romaine lettuce with parmesan and croutons</p>
        <span class="price">$12.99</span>
      </div>
    </div>
  </div>
</section>

```

```

<section id="gallery">
  <div class="container">
    <h2>Gallery</h2>
    <div class="slider">
      <div class="slides">
        <div class="slide">
      <button class="prev" onclick="prevSlide()">&#10094;</button>
      <button class="next" onclick="nextSlide()">&#10095;</button>
    </div>
  </div>
</section>

<section id="contact">
  <div class="container">
    <h2>Contact Us</h2>
    <div class="contact-content">
      <div class="contact-info">
        <p>Visit us or get in touch!</p>
        <p>Address: 123 Main Street, City, State 12345</p>
        <p>Phone: (123) 456-7890</p>
        <p>Email: info@deliciousbites.com</p>
      </div>
    </div>
  </div>
</section>

```

Style.css

```

/* Reset and Base Styles */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Arial', sans-serif;
  line-height: 1.6;
  color: #333;
  scroll-behavior: smooth;
}

.container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 0 20px;
}

/* Navigation */
#nav {
  position: fixed;
  top: 0;
  width: 100%;
  background-color: #fff;
  box-shadow: 0 2px 5px #000;
  z-index: 1000;
}

```

```

.nav-container {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 1rem 2rem;
}

.logo {
  font-size: 1.5rem;
  font-weight: bold;
  color: #d2691e;
}

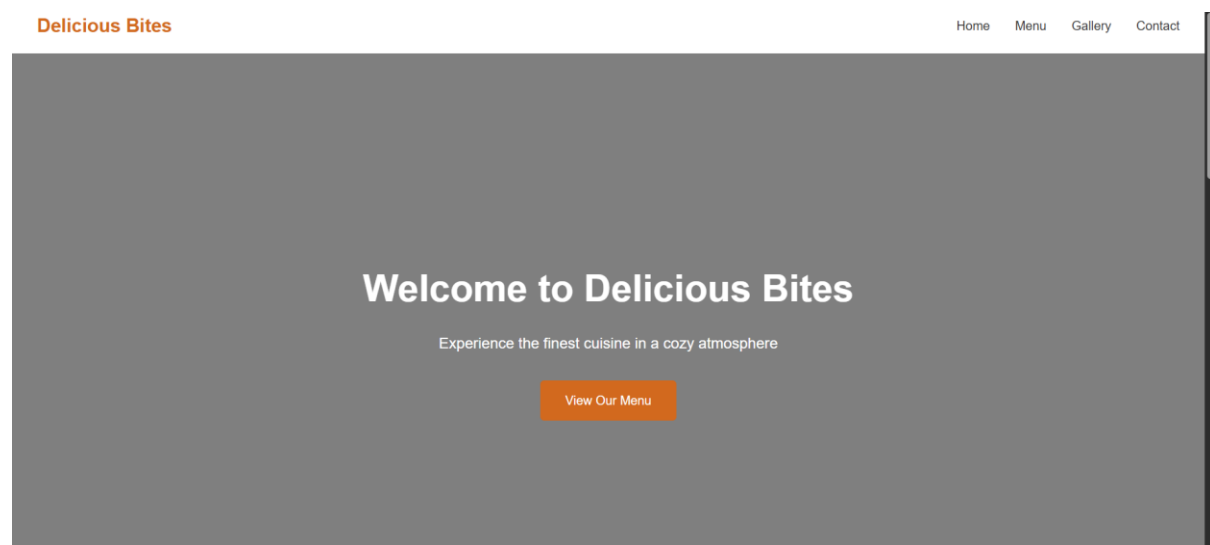
.nav-links {
  display: flex;
  list-style: none;
}

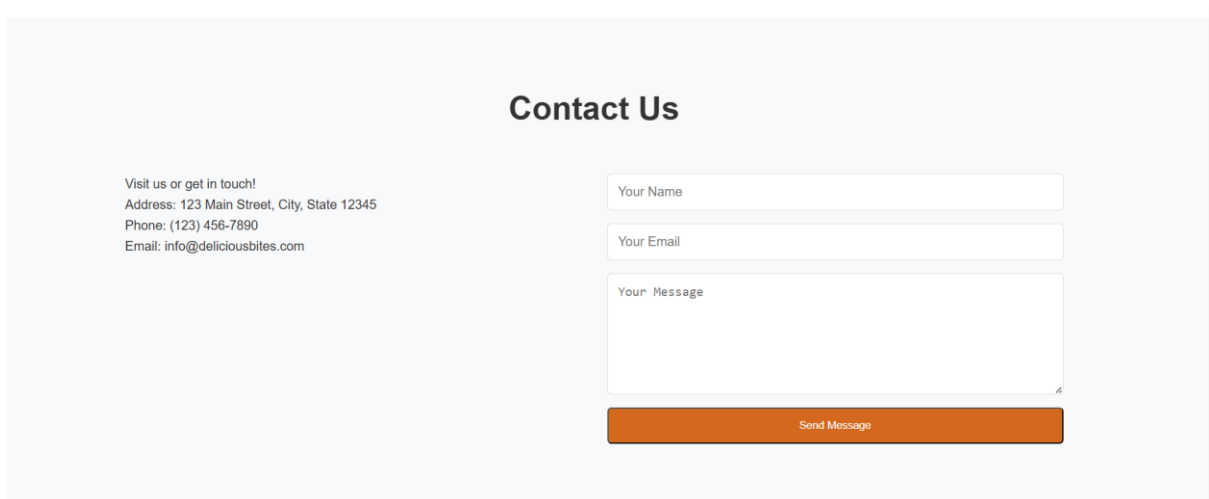
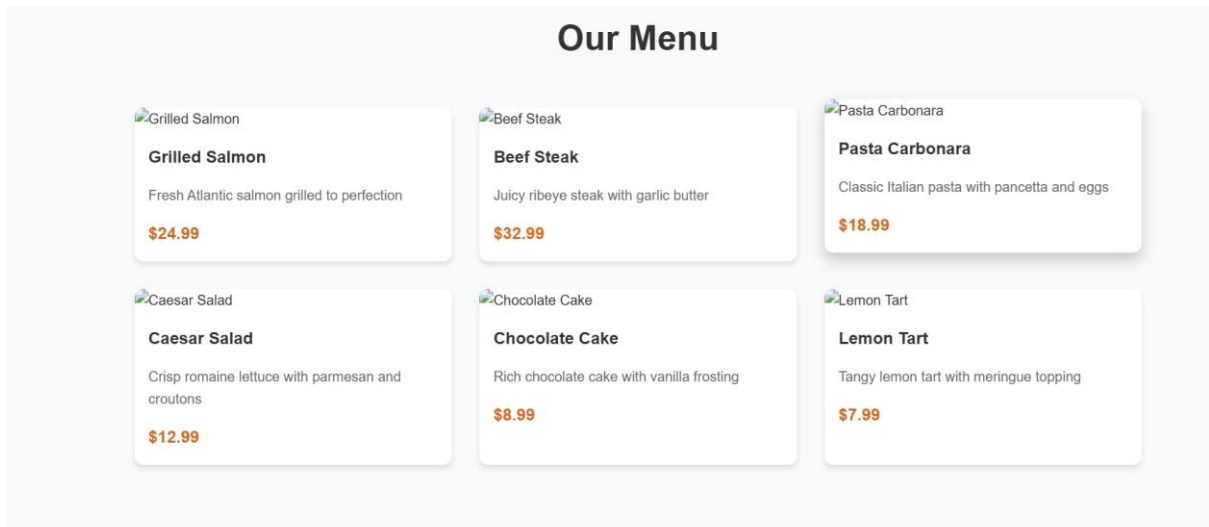
.nav-links li {
  margin-left: 2rem;
}

.nav-links a {
  text-decoration: none;
  color: #333;
  font-weight: 500;
  transition: color 0.3s;
}

```

Output:





Code Explanation:

The restaurant landing page is designed to present the restaurant's menu, images, and contact details using **HTML, CSS, and JavaScript**.

HTML Structure

The HTML file organizes the layout of the page.

- The `<nav>` element contains navigation links: **Home, Menu, Gallery, and Contact**.
- The **Home section** acts as a hero section displaying the restaurant name and tagline.
- The **Menu section** uses card layouts to display food items including **image, name, and price**.
- The **Gallery section** displays restaurant images inside a slider container.

CSS Styling

CSS is used to style the landing page.

- **Flexbox or Grid layout** is used to arrange menu items in cards.
- Each card includes padding, shadows, and border-radius for a modern design.
- Hover animations highlight menu items when users move the mouse over them.
- Media queries ensure the website is responsive across different screen sizes.

JavaScript Functionality

JavaScript is used to implement the **image slider in the gallery section**.

- A slider function cycles through images automatically.
- Buttons allow users to navigate between images using **next and previous controls**.
- Smooth transitions improve the user experience.

This creates a **visually attractive restaurant landing page with interactive gallery functionality**.

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

- o Name
- o Email
- o Phone Number
- o Department
- o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

- o Email format validation
- o Phone number length validation

o Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions

Prompt used:

Generate an event registration form using HTML, CSS, and JavaScript.

Form fields: Name, Email, Phone Number, Department, and Event Selection (dropdown).

- Style the form with modern CSS for readability and accessibility.
- Add real-time JavaScript validation:
 - Email format validation
 - Phone number should be exactly 10 digits
 - All fields must be required
- Show validation error messages below the fields.
- Prevent form submission if inputs are invalid.
- Ensure the form layout is responsive and mobile friendly.

AI Generated Code:

Index.html


```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Event Registration Form</title>
  <link rel="stylesheet" href="event-form.css">
</head>
<body>
  <div class="container">
    <h1>Event Registration</h1>
    <form id="registrationForm">
      <div class="form-group">
        <label for="name">Name:</label>
        <input type="text" id="name" name="name" required>
        <span class="error" id="nameError"></span>
      </div>

      <div class="form-group">
        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>
        <span class="error" id="emailError"></span>
      </div>

      <div class="form-group">
        <label for="phone">Phone Number:</label>
        <input type="tel" id="phone" name="phone" required>
        <span class="error" id="phoneError"></span>
      </div>

```

```

      <div class="form-group">
        <label for="department">Department:</label>
        <input type="text" id="department" name="department" required>
        <span class="error" id="departmentError"></span>
      </div>

      <div class="form-group">
        <label for="event">Event Selection:</label>
        <select id="event" name="event" required>
          <option value="">Select an event</option>
          <option value="conference">Tech Conference</option>
          <option value="workshop">Coding Workshop</option>
          <option value="seminar">AI Seminar</option>
          <option value="networking">Networking Event</option>
        </select>
        <span class="error" id="eventError"></span>
      </div>

      <button type="submit" class="submit-btn">Register</button>
    </form>
  </div>
  <script src="event-form.js"></script>
</body>
</html>

```

```

h1 {
  text-align: center;
  margin-bottom: 2rem;
  color: #2c3e50;
}

/* Form Styles */
.form-group {
  margin-bottom: 1.5rem;
}

label {
  display: block;
  margin-bottom: 0.5rem;
  font-weight: bold;
  color: #555;
}

input, select {
  width: 100%;
  padding: 0.8rem;
  border: 2px solid #ddd;
  border-radius: 5px;
  font-size: 1rem;
  transition: border-color 0.3s, box-shadow 0.3s;
}

input:focus, select:focus {
  outline: none;

```

```

/* Reset and Base Styles */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Arial', sans-serif;
  line-height: 1.6;
  color: #333;
  background-color: #f4f4f4;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  padding: 20px;
}

.container {
  background-color: #fff;
  padding: 2rem;
  border-radius: 10px;
  box-shadow: 0 4px 6px #0000000a;
  max-width: 500px;
  width: 100%;
}

h1 {

```

```

.submit-btn {
  width: 100%;
  padding: 0.8rem;
  background-color: #3498db;
  color: white;
  border: none;
  border-radius: 5px;
  font-size: 1rem;
  font-weight: bold;
  cursor: pointer;
  transition: background-color 0.3s, transform 0.2s;
}

.submit-btn:hover {
  background-color: #2980b9;
  transform: translateY(-2px);
}

.submit-btn:active {
  transform: translateY(0);
}

/* Responsive Design */
@media (max-width: 600px) {
  .container {
    padding: 1.5rem;
    margin: 10px;
  }
}

```

Output:

Event Registration

Name:

Please fill out this field.

Email:

Phone Number:

Department:

Event Selection:

Select an event

▼

Code explanation:

The event registration form allows users to register for an event while ensuring correct data entry using **HTML, CSS, and JavaScript validation**.

HTML Structure

The HTML form collects user information.

The form includes the following input fields:

- Name
- Email
- Phone Number
- Department
- Event Selection (Dropdown)

Each field uses <input> or <select> elements and includes **labels for accessibility**.

CSS Styling

CSS improves the visual appearance of the form.

- The form container is centered on the page with padding and border styling.
- Input fields are styled with consistent spacing, font size, and border design.
- Buttons are styled with hover effects for better user interaction.

JavaScript Validation

JavaScript ensures the user enters valid information.

The validation includes:

- **Email validation** using pattern matching to confirm a proper email format.
- **Phone number validation** to ensure exactly **10 digits** are entered.
- **Required field validation** to prevent empty submissions.

If incorrect data is entered, **error messages are displayed** and the form submission is prevented.

This ensures **accurate and valid user registration data**.

Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
 - o Cart item counter at the top-right corner
 - o Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

AI Generated Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple E-commerce Product Page</title>
  <link rel="stylesheet" href="ecommerce.css">
</head>
<body>
  <header>
    <h1>Simple Shop</h1>
    <div class="cart">
      <span class="cart-icon">🛒</span>
      <span class="cart-count" id="cartCount">0</span>
    </div>
  </header>

  <main>
    <div class="products-grid">
      <div class="product-card">
        
        <h3>Wireless Headphones</h3>
        <p class="price">$49.99</p>
        <button class="add-to-cart" data-product="1">Add to Cart</button>
      </div>
      <div class="product-card">
        
        <h3>Smart Watch</h3>

```

```

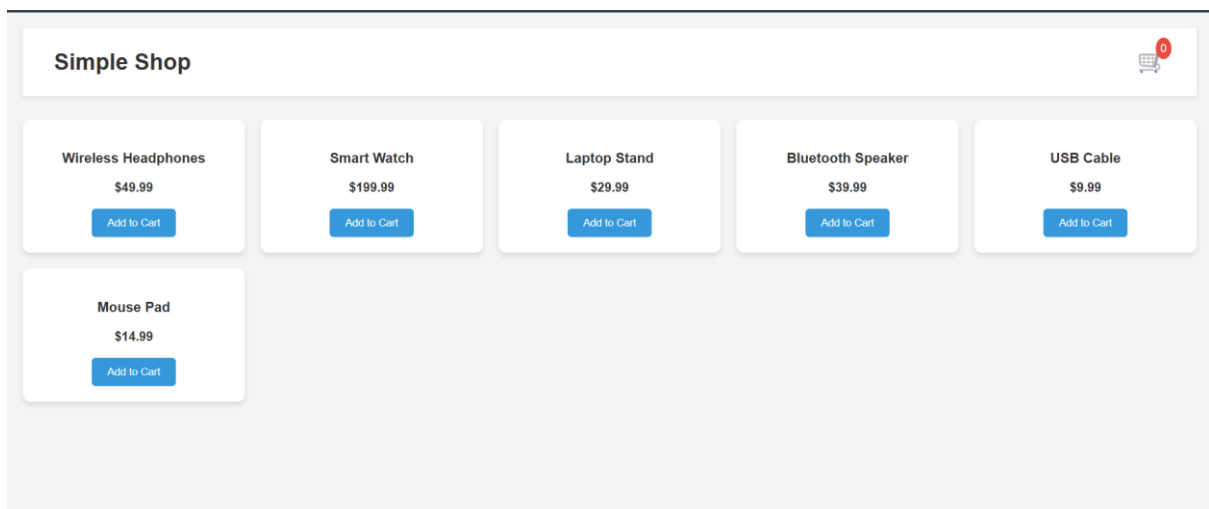
JS ecommerce.js > ...
1 document.addEventListener('DOMContentLoaded', () => {
2     const cartCountEl = document.getElementById('cartCount');
3     let cartCount = 0;
4
5     function updateCartCount() {
6         cartCountEl.textContent = cartCount;
7     }
8
9     function addToCart(event) {
10        const button = event.target;
11        cartCount++;
12        updateCartCount();
13
14        // Visual feedback on button
15        button.textContent = 'Added';
16        button.disabled = true;
17        button.style.backgroundColor = '#2ecc71';
18
19        // small notification (optional)
20        const notification = document.createElement('div');
21        notification.className = 'notification';
22        notification.textContent = 'Product added to cart!';
23        document.body.appendChild(notification);
24        setTimeout(() => {
25            notification.remove();
26        }, 2000);
27    }
28
29    document.querySelectorAll('.add-to-cart').forEach(btn => {

```

```

# restaurant.css # event-form.css X ecommerce.html Release Notes: 1.111.0 JS ecommerce.js # ecommerce.css et ...
# event-form.css > {} @media (max-width: 600px)
1 /* Reset and Base Styles */
2 * {
3     margin: 0;
4     padding: 0;
5     box-sizing: border-box;
6 }
7
8 body {
9     font-family: 'Arial', sans-serif;
10    line-height: 1.6;
11    color: #333;
12    background-color: #f4f4f4;
13    display: flex;
14    justify-content: center;
15    align-items: center;
16    min-height: 100vh;
17    padding: 20px;
18 }
19
20 .container {
21     background-color: #fff;
22     padding: 2rem;
23     border-radius: 10px;
24     box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
25     max-width: 500px;
26     width: 100%;
27 }
28
29 h1 {

```



Task 5: Responsive Web Page Layout

Code Explanation

The responsive webpage layout demonstrates the basic structure of a webpage using **HTML** and **CSS**.

HTML Structure

The webpage contains three main sections:

1. **Header**
 - Displays the website title and navigation links.
2. **Main Content Area**
 - Contains placeholder text, images, or articles.
3. **Footer**
 - Displays copyright information or contact details.

Semantic HTML tags such as `<header>`, `<main>`, and `<footer>` improve readability and accessibility.

CSS Styling

CSS is used to arrange the layout.

- **Flexbox or Grid layout** organizes content properly across the page.
- The header and footer are styled with background colors and padding.
- The main section contains text and images aligned neatly.

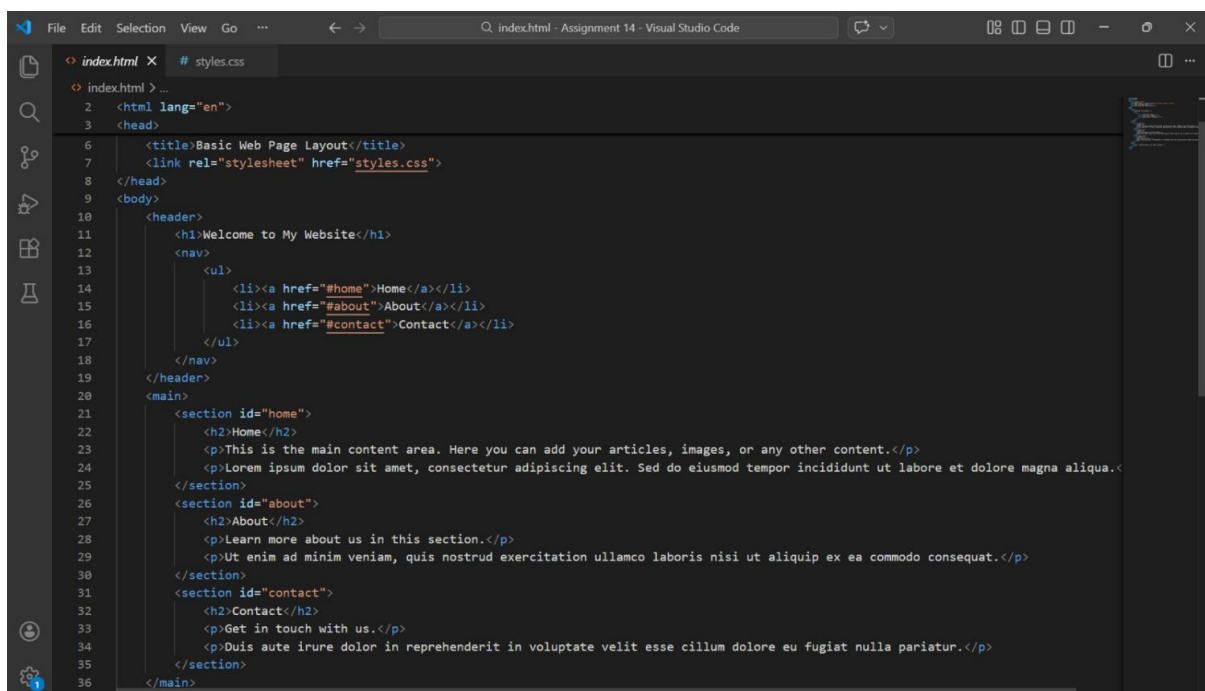
Responsive Design

Media queries are used to make the layout responsive.

- On **desktop screens**, content is displayed in wider layouts.
- On **tablets**, elements adjust to medium screen sizes.
- On **mobile devices**, elements stack vertically for better readability.

This ensures the webpage **adapts smoothly to different screen sizes**.

AI Generated Code –

A screenshot of the Visual Studio Code editor interface. The top bar shows the file name 'index.html - Assignment 14 - Visual Studio Code'. The editor window displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Basic Web Page Layout</title>
6   <link rel="stylesheet" href="styles.css">
7 </head>
8 <body>
9   <header>
10    <h1>Welcome to My Website</h1>
11    <nav>
12      <ul>
13        <li><a href="#home">Home</a></li>
14        <li><a href="#about">About</a></li>
15        <li><a href="#contact">Contact</a></li>
16      </ul>
17    </nav>
18  </header>
19  <main>
20    <section id="home">
21      <h2>Home</h2>
22      <p>This is the main content area. Here you can add your articles, images, or any other content.</p>
23      <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
24    </section>
25    <section id="about">
26      <h2>About</h2>
27      <p>Learn more about us in this section.</p>
28      <p>Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.</p>
29    </section>
30    <section id="contact">
31      <h2>Contact</h2>
32      <p>Get in touch with us.</p>
33      <p>Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.</p>
34    </section>
35  </main>
36 </body>
37 </html>
```

The code is syntax-highlighted, and the editor includes a sidebar on the left with icons for Explorer, Search, Source Control, and Run and Debug. The right sidebar shows the 'Styles' panel, which is currently empty.


```
index.html # styles.css X
# styles.css > % header
1 /* Reset and base styles */
2 * {
3     margin: 0;
4     padding: 0;
5     box-sizing: border-box;
6 }
7
8 body {
9     font-family: Arial, sans-serif;
10    line-height: 1.6;
11    color: #333;
12    background-color: #f4f4f4;
13 }
14
15 /* Header styles */
16 header {
17     background-color: #333;
18     color: #fff;
19     padding: 1rem;
20     text-align: center;
21 }
22
23 header h1 {
24     margin-bottom: 0.5rem;
25 }
26
27 nav ul {
28     list-style: none;
29     display: flex;
30     justify-content: center;
31     gap: 2rem;
32 }
33
34 nav a {
```

